

n8n Workshop Documentation

Workflow Automation & AI Agents

This document outlines the structure and curriculum for a 2-hour technical workshop designed to introduce participants to n8n, a powerful low-code tool for building AI-powered workflows and automating repetitive tasks.

1. Event Overview

- Total Duration: 2 Hours (120 Minutes)
- Target Audience: All Department Students
- Objective: To equip attendees with the skills to build autonomous AI workflows and automate business logic with minimal coding.

2. Learning Objectives

By the end of this session, participants will be able to:

1. Navigate the n8n interface and understand core concepts like nodes and workflows.
2. Connect disparate services (e.g., Gmail, Telegram, AI models) to automate data flow.
3. Implement AI agents within n8n to handle complex, non-linear tasks.

3. Workshop Agenda

Time Segment	Topic	Description
00:00 - 00:10	Intro to Automation	Defining workflow automation and an

Time Segment	Topic	Description
		overview of n8n vs. other tools.
00:10 - 00:20	Core n8n Mechanics	Understanding Nodes, Triggers, and the Visual Canvas.
00:20 - 00:45	Hands-on: Gmail-to-Telegram Assistant	Building a Telegram Gmail Assistant that fetches emails from Gmail and sends them to users on Telegram.
00:45 - 01:15	AI Agent Integration	Introduction to AI Nodes and building a basic AI chatbot in n8n.
01:15 - 02:00	RAG Implementation	Building a RAG application to retrieve relevant information from documents and generate context-aware AI responses.

4. Technical Modules

4.1 Understanding Nodes and Workflows

Workflows are sequences of nodes connected by lines. Each node performs a specific action, such as fetching data from a database or sending an email.

- **Trigger Nodes:** Start the workflow (e.g., a new email arrives or a schedule is met).
- **Action Nodes:** Execute tasks (e.g., GPT-4 processing, updating a Google Sheet).

4.2 Real-World Case Study: Telegram Gmail Assistant

- Participants will build a real-world automation workflow that connects Telegram with Gmail. The system allows users to interact with their Gmail account directly through a Telegram bot.

The bot will automatically:

- Fetch Gmail emails based on the user's Telegram request.
- Display email details such as sender, subject, and message content directly in Telegram.
- Search and filter emails using commands such as unread, today, sender, or subject.
- Send email notifications to Telegram when new emails arrive.

4.3 Introduction to AI Agents with RAG

- The workshop will demonstrate how to combine AI agents with Retrieval-Augmented Generation (RAG) to build intelligent automation workflows using n8n. Participants will learn how an AI agent can understand user queries, retrieve relevant information from a knowledge base, and use connected tools to perform tasks.
- This includes:
- Connecting an LLM (Large Language Model) to an n8n AI Agent.
- Building a RAG knowledge base using documents, PDFs, or other data sources.
- Connecting a vector database to retrieve relevant information.
- Giving the AI Agent tools/skills such as web search, Gmail, Telegram, or database access.
- Using retrieved context to generate accurate and context-aware responses.
- Adding memory so the AI agent can maintain context across conversations.
- Creating an end-to-end AI agent + RAG workflow for real-world use cases.